

Thesis Proposal

Pablo Ariel Rodriguez
Candidate in Electronics Engineering
Universidad de Buenos Aires, Facultad de Ingeniería

Supervisor: Ing. Ricardo Veiga

Date: December 2025

Reconstruction of Volatility Surfaces Using Transformer-Based Autoencoders with No-Arbitrage Constraints

1. Background and Motivation

In quantitative finance, **options** are financial derivatives whose value depends on the future price of an underlying asset (such as a stock, exchange rate, or commodity). A European option is characterized by a **strike price** (price level), a **maturity** (future time), and a payoff type (call or put).

A cornerstone of option pricing theory is the **Black–Scholes model**, which assumes that asset prices follow a log-normal diffusion with constant volatility. Under these assumptions, option prices admit a closed-form solution. While the model is analytically convenient and widely used, its core assumption of constant volatility is strongly violated in real markets.

In practice, market option prices are converted into **implied volatilities**: the volatility value that, when inserted into the Black–Scholes formula, reproduces the observed price. Empirically, implied volatility is not constant but varies systematically with:

- the strike (or log-moneyness)
- the time to maturity

This leads to the concept of a **volatility surface**, a two-dimensional function

$$\sigma = \sigma(\textit{strike}, \textit{maturity})$$

which summarizes market expectations of future uncertainty. Volatility surfaces exhibit strong structure:

- they are smooth but non-linear
- they are constrained by financial no-arbitrage conditions
- they are effectively low-dimensional despite being defined over a 2D domain

- they evolve slowly over time but must be reconstructed frequently

From a machine learning perspective, volatility surfaces resemble structured fields or images that lie on a constrained manifold, making them a natural target for representation learning.

2. The Practical Problem

In real markets, option quotes are:

- sparse and irregular across strikes and maturities
- noisy and sometimes internally inconsistent
- incomplete in less liquid regions

However, downstream tasks such as pricing, hedging, and risk management require a **complete, smooth, arbitrage-free volatility surface**.

The core practical problem can therefore be stated as:

Reconstructing a full implied volatility surface from sparse and noisy observations while enforcing no-arbitrage constraints.

Traditional solutions rely on parametric models or calibrated stochastic processes. While theoretically grounded, these approaches:

- impose strong functional assumptions
- require delicate calibration
- do not explicitly exploit data-driven low-dimensional structure

3. Objective of the Thesis

The objective of this thesis is to **frame volatility surface reconstruction as a representation learning problem** and to solve it using **transformer-based autoencoders** augmented with **no-arbitrage constraints**.

More precisely, the thesis investigates the following question:

Can a transformer-based autoencoder learn a compact latent representation of volatility surfaces that enables accurate reconstruction from sparse observations, while respecting financial no-arbitrage conditions?

The key idea is not to replace financial structure, but to combine:

- data-driven learning of surface geometry

- explicit financial constraints ensuring admissibility

4. Methodology Overview

4.1 Data Representation

Each volatility surface is represented on a discretized grid over:

- log-moneyness (strike normalized by spot price)
- time to maturity

Observed market data correspond to **partial observations** of this grid. The reconstruction task is therefore analogous to **masked field or image completion**.

4.2 Transformer-Based Autoencoder

The proposed model is a **transformer-based autoencoder**:

- **Encoder:**
 - Takes sparse or masked volatility observations,
 - Uses attention mechanisms to capture global dependencies across strikes and maturities
 - Produces a low-dimensional latent representation
- **Decoder:**
 - Reconstructs the full volatility surface from the latent code
 - Outputs a dense and smooth surface

Transformers are particularly suitable because:

- volatility surfaces exhibit long-range, non-local structure,
- correlations across distant strikes and maturities are important,
- attention mechanisms naturally handle irregular and masked inputs.

4.3 No-Arbitrage Constraints

Not all volatility surfaces are financially admissible. Absence of static arbitrage implies constraints such as:

- monotonicity of option prices with respect to strike
- convexity of option prices in strike
- monotonicity with respect to maturity

These constraints translate into inequality conditions on option prices and, indirectly, on implied volatilities.

The thesis explores methods to incorporate these constraints into learning, including:

- penalty terms in the reconstruction loss
- constraint-aware regularization
- projection or filtering of decoded surfaces

This introduces a **constraint-aware / physics-informed learning** component.

5. Expected Contributions

The expected contributions of this thesis are:

1. A **data-driven framework** for volatility surface reconstruction using transformer autoencoders.
2. A **low-dimensional latent representation** capturing the intrinsic geometry of volatility surfaces.
3. Integration of **no-arbitrage constraints** into a modern deep learning architecture.
4. An **empirical evaluation** on synthetic and/or real market data, comparing reconstruction accuracy and arbitrage violations against classical approaches.

6. Related Work

6.1 Classical Volatility Surface Models

Traditional approaches rely on parametric or stochastic models:

- Black, F. and Scholes, M. (1973). *The Pricing of Options and Corporate Liabilities*.
- Heston, S. (1993). *A Closed-Form Solution for Options with Stochastic Volatility*.
- Gatheral, J. (2006). *The Volatility Surface: A Practitioner's Guide*.

These models provide interpretability but are often rigid and difficult to calibrate robustly.

6.2 Autoencoders and Latent Representations

Several works explicitly use autoencoders to model volatility surfaces:

- Bloch, D. (2021). *Deep Learning for Volatility Surfaces*.
- Horvath, B., Muguruza, A., and Tomas, M. (2021). *Deep Learning Volatility*.

- Bergeron, M., Fung, N., Hull, J., & Poulos, Z. (2021). *Variational Autoencoders: A Hands-Off Approach to Volatility*.
- Feugang Nteumagné, B., Azemtsa Donfack, H., & Wafo Soh, C. (2025). *Variational Autoencoders for Completing the Volatility Surfaces*. *Journal of Risk and Financial Management*, **18**(5), 23